



Solution Set

Question 1

A portfolio PnL is

$$\Pi = 4X - 2Y$$

with

$$\text{Var}(X) = 9, \quad \text{Var}(Y) = 16, \quad \text{Corr}(X, Y) = 0.35$$

Compute

$$\text{Var}(\Pi)$$

Answer key

First compute covariance:

$$\text{Cov}(X, Y) = 0.35(3)(4) = 4.2$$

Then

$$\begin{aligned} \text{Var}(4X - 2Y) &= 16\text{Var}(X) + 4\text{Var}(Y) + 2(4)(-2)\text{Cov}(X, Y) \\ &= 16(9) + 4(16) - 16(4.2) = 140.8 \end{aligned}$$

Question 2

Prove that independence implies zero covariance when second moments exist.

Answer key

By definition,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Independence gives

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

so covariance is zero. The converse is false because zero covariance removes only linear dependence.

Question 3

Give a concrete example where covariance is zero but variables are dependent.

Answer key

Let X be equally likely to be $-1, 0$, or 1 , and let $Y = X^2$. Then Y is determined by X , so they are dependent. But

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0$$

so

$$\text{Cov}(X, Y) = 0$$

This is a standard interview trap.

Question 4

Derive the minimum-variance hedge ratio by projecting portfolio PnL onto hedge PnL.

Answer key

Minimize

$$\mathbb{E}[(P - hH)^2]$$

Differentiate with respect to h:

$$\frac{d}{dh}\mathbb{E}[(P - hH)^2] = -2\mathbb{E}[H(P - hH)]$$

Set equal to zero:

$$\mathbb{E}[PH] - h\mathbb{E}[H^2] = 0$$

If variables are centered,

$$h^* = \frac{\text{Cov}(P, H)}{\text{Var}(H)}$$

Question 5

State the law of large numbers and explain why it does not eliminate tail risk.

Answer key

The law of large numbers says sample averages converge to expectations under suitable assumptions. It does not say rare events cannot occur, nor does it guarantee accurate tail quantile estimation from small samples. Tail risk estimation often needs far more data than mean estimation.

Question 6

Use the central limit theorem to approximate the distribution of the average of n independent daily returns with mean μ and variance σ^2 .

Answer key

For independent finite-variance returns R_i ,

$$\bar{R}_n = \frac{1}{n} \sum_{i=1}^n R_i$$

is approximately normal for large n:

$$\bar{R}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

The approximation concerns averages, not necessarily the raw one-day return distribution.

Question 7

A two-asset portfolio has weights 0.7 and 0.3, volatilities 20 percent and 35 percent, and correlation -0.25. Compute portfolio volatility.

Answer key

Variance is

$$\begin{aligned}\sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2 \\ &= 0.49(0.04) + 0.09(0.1225) + 2(0.7)(0.3)(-0.25)(0.2)(0.35) \\ &= 0.0196 + 0.011025 - 0.00735 = 0.023275 \\ \sigma_p &= \sqrt{0.023275} = 15.26\%\end{aligned}$$

Question 8

Explain why square-root-of-time VaR scaling can fail.

Answer key

Square-root scaling assumes independent identically distributed returns with stable variance and roughly linear aggregation. It can fail under autocorrelation, volatility clustering, jumps, changing positions, nonlinear derivatives, and heavy tails. For options, gamma alone can break simple scaling.

Question 9

Prove the variance decomposition

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$$

Answer key

Write

$$X - \mathbb{E}[X] = \{X - \mathbb{E}[X | Y]\} + \{\mathbb{E}[X | Y] - \mathbb{E}[X]\}$$

Square and take expectations. The cross term is zero because

$$\mathbb{E}[X - \mathbb{E}[X | Y] | Y] = 0$$

Thus

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$$

Question 10

What is convergence in distribution, and why is it weaker than convergence in probability?

Answer key

Convergence in distribution means the CDFs converge at continuity points of the limiting CDF. It does not require the random variables to become close on the same probability space. Convergence in probability is stronger because it requires

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$$

for every epsilon > 0.

Question 11

Explain Jensen's inequality and give a risk-management interpretation.

Answer key

For convex phi,

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$$

Convex losses penalize dispersion. For example, if loss utility is convex, uncertainty increases expected disutility even when expected PnL is unchanged.

Question 12

A sample correlation matrix has a negative eigenvalue. What does this imply and how would you respond?

Answer key

A valid correlation matrix must be positive semidefinite. A negative eigenvalue means some portfolio has negative computed variance. The response is to inspect data alignment, missing values, estimation noise, and then repair or shrink the matrix before using it in VaR or optimization.